Information Systems Applications

DATA MINING

Data Mining

- Data mining is the process of analysing data from different perspectives and summarise it into information that can used to increase revenues or decrease costs.
- Similarly, Data mining is the process of analysing data to extract information not offered by the raw data alone.
- Data mining also includes establishing relationships and finding patterns, anomalies, and correlations to tackle issues, creating actionable information in the process. It is a wide-ranging and varied process that includes many different components, some of which are even confused with data mining itself.

Data Mining Steps

1. **Understanding and Guaging Data:** The first step in the data mining process is knowing your data. You must thoroughly understand the data to identify its characteristics, quality, and relevance. You must also gauge its structure, volume, and nature and determine its relevance to the business objectives.

Data Mining Steps contd/

2. **Data Preparation:** Data preparation involves preparing the data for mining by cleaning, transforming, and selecting relevant data. Here's all about it in detail.
   i. Data Cleaning: In this step, you should remove noise, handle missing values, and correct errors.
   ii. Data Integration: This step includes combining data from different sources into a coherent data set.
   iii. Data Transformation: Normalize or aggregate data to ensure consistency and improve mining results.
   iv. Data Reduction: Reduce the data volume by selecting only relevant features, creating new features, or sampling.

Data Mining Steps contd/

3. **Data Selection:** The next step in the overall data mining process is data selection. You must define criteria for selecting relevant data and extract the appropriate subset of data for mining.
4. **Data Mining**: Next you extract patterns and insights from the prepared data. You should choose appropriate data mining techniques, such as classification, clustering, and regression, and apply them to the data. Once you have done this, perform iterative testing and validation to refine the mining process.
5. **Pattern Evaluation and Presentation:** Visualize patterns and insights using charts, graphs, and dashboards, and prepare reports to communicate your findings. And then present the mined knowledge in an actionable format.

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5. **Pattern Evaluation and Presentation:** Visualize patterns and insights using charts, graphs, and dashboards, and prepare reports to communicate your findings. And then present the mined knowledge in an actionable format. (Earn a brownie point by also interpreting your findings in the context of business objectives.)

Real Worl examples where Data Mining is applied

1. **Shopping Market Analysis**
   - In the shopping market, there is a big quantity of data, and the user must manage enormous amounts of data using various patterns. To do the study, market basket analysis is a modeling approach.

   - Market basket analysis is basically a modeling approach that is based on the notion that if you purchase one set of products, you're more likely to purchase another set of items.

Real world examples where data mining is applied

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2. **Weather Forecasting Analysis**

   For prediction, weather forecasting systems rely on massive amounts of historical data. Because massive amounts of data are being processed, the appropriate data mining approach must be used.

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Real world examples where data mining is applied

3. **Stock Market Analysis**

   In the stock market, there is a massive amount of data to be analyzed. As a result, data mining techniques are utilized to model such data in order to do the analysis.

4. **Intrusion Detection**

   Well, data mining can assist to enhance intrusion detection by focusing on anomaly detection. It assists an analyst in distinguishing between unusual network activity and normal network activity.

Real world examples where data mining is applied

5. **Fraud Detection**

   Traditional techniques of fraud detection are time-consuming and difficult due to the amount of data. Data mining aids in the discovery of relevant patterns and the transformation of data into information.

6. **Surveillance**

   Well, video surveillance is utilized practically everywhere in everyday life for security perception. Because we must deal with a huge volume of acquired data, data mining is employed in video surveillance.

Real world examples where data mining is applied

7. **Financial Banking**

   With each new transaction in computerized banking, a massive amount of data is expected to be created.

   By identifying patterns, causalities, and correlations in corporate data, data mining may help solve business challenges in banking and finance.

Benefits of Data Mining

1. It helps companies gather reliable information and businesses make informed decisions
2. It's an efficient, cost-effective solution compared to other data applications
3. It helps businesses make profitable production and operational adjustments
4. It helps detect credit risks and fraud
5. It helps data scientists easily analyze enormous amounts of data quickly
6. It helps data scientists quickly initiate automated predictions of behaviors and trends and discover hidden patterns

https://www.simplilearn.com/what-is-data-mining-article#what_are_the_benefits_of_data_mining